

System Analysis Task

1 - Library Management System

Step 1 - Identify the Entities

Entity Name: Book

Property Name	Data Type	Source Tag
bookId	int	System Generated
title	string	User Input
author	string	User Input
genre	string	User Input
totalCopies	int	User Input
availableCopies	int	Calculated
borrowCount	int	System Generated

Entity Name: Member

Property Name	Data Type	Source Tag
memberId	int	System Generated
name	string	User Input
email	string	User Input
phone	string	User Input
registrationDate	DateTime	User Input /System Generated

Entity Name: Loan

Property Name	Data Type	Source Tag
loanId	int	System Generated
memberId	int	User Input
bookId	int	From List
loanDate	DateTime	User Input /System Generated
dueDate	DateTime	Calculated
returnDate	DateTime	User Input
fineAmount	decimal	Calculated
status	string	Default Value ("Active")

Entity Name: Reservation

Property Name	Data Type	Source Tag
reservationId	int	System Generated
memberId	int	User Input
bookId	int	From List
reservationDate	DateTime	System Generated
isNotified	bool	Default Value (false)

Step 2 - Entity Relationships

- A member registers in the system.
- A book is stored with its details and available copies.
- A loan is created when a member borrows a book, reducing availablecopies.
- If overdue, the system calculates fines.
- A reservation is created if a book is unavailable, linking member and book.
- When a book is returned, the loan closes, availablecopies increases, and reservations may trigger notifications.

Step 3 – Services

	Service Name	What it does
1	RegisterMember	Registers a new library member
2	AddBook	Adds a new book to the catalog
3	BorrowBook	Creates a loan record for a member
4	ReturnBook	Marks loan as returned and updates availability
5	CalculateFine	Computes overdue fines
6	ReserveBook	Allows member to reserve unavailable book
7	NotifyReservation	Notifies member when reserved book is available
8	ViewActiveLoans	Displays all current loans
9	ViewOverdueLoans	Lists overdue loans sorted by lateness
10	ReportMostBorrowed	Generates report of most borrowed books

2 - Hotel Management System

Step 1 - Identify the Entities

Entity Name: Guest

Property Name	Data Type	Source Tag
guestId	int	System Generated
guestName	string	User Input
guestPhone	string	User Input
guestEmail	string	User Input
guestNationality	string	User Input
registrationDate	DateTime	System Generated

Entity Name: Room

Property Name	Data Type	Source Tag
roomId	int	System Generated
roomNumber	string	User Input
roomType	string	User Input
floorNumber	int	User Input
nightlyPrice	decimal	User Input
isAvailable	bool	Default Value (true)

Entity Name: Booking

Property Name	Data Type	Source Tag
bookingId	int	System Generated
guestId	int	User Input
roomId	int	From List/ User Input
checkInDate	DateTime	User Input
checkOutDate	DateTime	User Input
numberOfNights	int	Calculated
totalPrice	decimal	Calculated
status	string	Default Value ("Booked")

Entity Name: CheckInRecord

Property Name	Data Type	Source Tag
recordId	int	System Generated
bookingId	int	User Input
guestId	int	Calculated
roomId	int	Calculated
actualCheckIn	DateTime	System Generated
actualCheckOut	DateTime	User Input
finalAmount	decimal	Calculated

Step 2 - Entity Relationships

- A Guest registers in the system.
- A Room is defined with type, price, and availability.
- A Booking links guest and room, setting dates and calculating total price.
- A CheckInRecord is created when guest checks in, marking room unavailable.
- On checkout, the record closes, finalAmount is calculated, and room becomes available again.
- Cancelling a booking frees the room immediately.

Step 3 - Services

	Service Name	What it does
1	RegisterGuest	Registers a new guest
2	AddRoom	Adds a new room to hotel system
3	MakeBooking	Creates a booking for a guest
4	CancelBooking	Cancels a booking before check-in
5	CheckInGuest	Creates a check-in record
6	CheckOutGuest	Completes stay, calculates final amount
7	CalculateTotalPrice	Computes total stay cost
8	ViewAvailableRooms	Lists all available rooms
9	ViewActiveBookings	Displays all current bookings